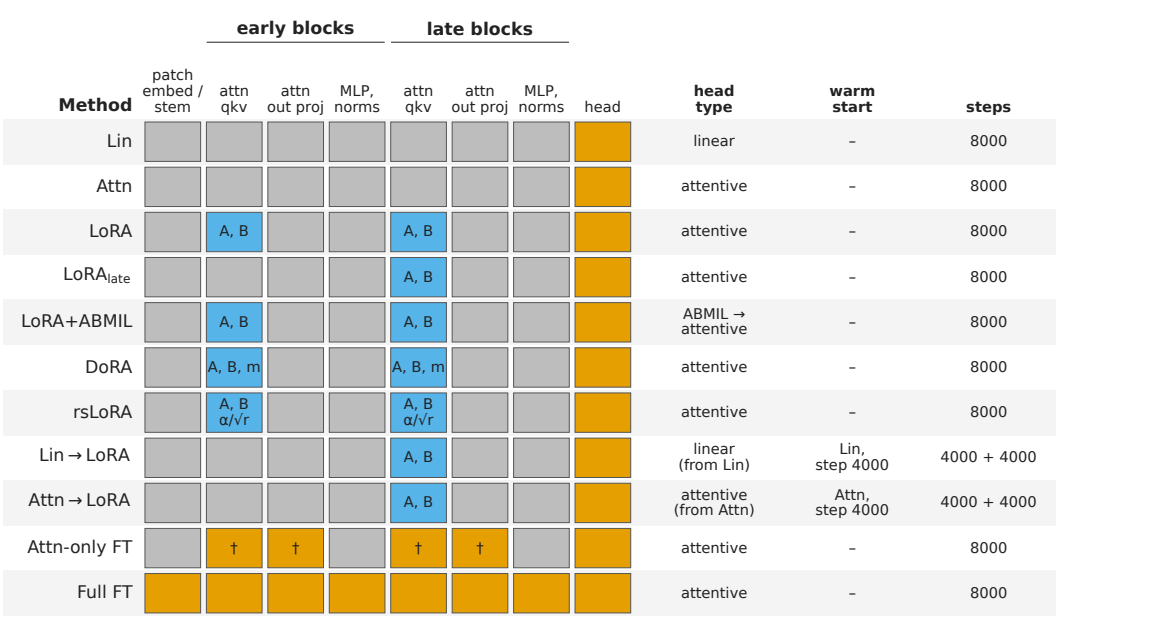

Late blocks per backbone
TAP-CT: blocks 9–12 of 12. SPECTRE: blocks 17–24 of 24 plus the 4 feature-combiner blocks. Pillar-0: stage 3 of 3 (8 of 12 multi-scale blocks, 3 scales each). VoCo: stage 4 of 4 (2 of 8 blocks).

LoRA target per backbone
Fused qkv Linear: TAP-CT, VoCo. Separate q and kv Linears: SPECTRE (backbone and combiner), Pillar-0. r 16, α 32, dropout 0.1 on the low-rank branch; scale α/r = 2; rsLoRA α/√r = 8.

DoRA
$W = m \cdot (W_0 + 2BA) / \|W_0 + 2BA\|$ per output row; m initialised to the row norms of W_0 and trained with A, B.

† Attn-only FT
Every parameter of each matched attention module: qkv (or q, kv), out proj, their biases, and for VoCo the relative-position bias table. MLPs, norms and the patch embedding stay frozen.

Warm start (two-stage)
Head (for VoCo also the running token standardiser) copied from the parent's step-4000 checkpoint of its 8000-step run; adapters start fresh (B = 0); new optimizer, 4000 steps.

Learning rate / weight decay
Lin 1e-3 / 0; Attn and all adapter methods 1e-4 / 0.01; Attn-only FT and Full FT backbone 2e-5 / 0.05, head 1e-4 / 0.01. Head dropout 0.2; batch 1.

frozen (pretrained weights) low-rank adapter, trained trained in full